

CSCE 632 ACCESSIBLE COMPUTING · MODULE 01 · LECTURE NOTES

Disability as a Design Failure

Where the problem is located, and why that decides everything else.

MODULE OUTCOMES

- State the medical and social models and their consequences.
- Explain why the location of the problem decides the engineering.
- State the curb-cut argument and its limits.
- Explain what this course means by a capability.
- Place this course relative to CSCE 671 and CSCE 679.

1 Two models

THE MEDICAL MODEL LOCATES DISABILITY IN THE PERSON AND THE SOCIAL MODEL LOCATES IT IN THE MISMATCH

The medical model says a person is disabled by their impairment — the impairment is the problem and it is located in them — so **the response is to treat the person where that is possible, and otherwise to accommodate them as an exception** to a system that remains as it was.

The social model says a person is disabled by an environment built for a narrower range of bodies than actually exists — the stairs disable the wheelchair user, not the wheelchair — so **the response is to fix the environment**.

And software is environment, almost entirely: there is no physical constraint forcing a button to be unreachable by keyboard or an image to be unlabelled — **which puts the defect squarely in the software and makes it an engineering problem with an owner, a severity, and a fix.**

So this course takes the social model as its working position, not as an ideological commitment but because it is the one that generates testable requirements — and a requirement you can test is what an engineering course needs (§3's capability framing is the direct consequence).

- **Under the medical model, an inaccessible interface is unfortunate** — and **the response is a separate accessible version**, requested individually and provided as an accommodation.
- **Under the social model it is a bug**, with a severity, an owner, a ticket, and a regression test — **which is the framing that actually gets things fixed**, because it routes the work through the process that fixes things.

- **And the separate version always loses — it lags the main product, it has fewer features, and it is the first thing cut when the schedule tightens — which is an observed pattern across two decades of text-only alternatives** rather than a prediction.
- **So the engineering consequence is: one artefact, built to work** — which is what the remaining twelve modules are about, and is also considerably cheaper than maintaining two.
- **And none of this is a claim that impairments are not real**, which the social model is sometimes accused of; **it is a claim about where the fixable part is**, and the fixable part is the one an engineer is responsible for.

2 The curb cut

DESIGNING FOR A CONSTRAINT IMPROVES THE DESIGN FOR EVERYBODY, AND THIS IS OBSERVED RATHER THAN HOPED

Pavement ramps were built for wheelchair users and are used by anybody with a pushchair, a suitcase, a delivery trolley, a bicycle, or a temporarily injured ankle — which is the original case and is where the name comes from.

And the software cases are stronger than the original: captions are used most by people who can hear (in noisy places, in quiet places, in a second language); **keyboard access is precisely what power users ask for**; and **clear, short language helps everybody who is tired or under time pressure** — which is everybody, sometimes.

So the accessibility argument does not require charity to work — it is a quality argument, and the quality argument is the one that survives a budget meeting, which Module 12 §2 treats as the practical fact it is.

But the honest limit, which this course states rather than eliding: not every accessibility requirement benefits everybody. A correct screen reader label on an icon benefits exactly the people who need it and nobody else, and it is still required. The curb-cut argument is a bonus, not the justification — and treating it as the justification leads to dropping the requirements that lack a mainstream beneficiary.

3 Capabilities

DESIGN AGAINST CAPABILITIES, NOT AGAINST DIAGNOSES

A diagnosis does not tell you what somebody can do — two people with the same recorded condition may have entirely different interaction needs, and in any case a diagnosis is private information you have no right to and will not have.

A capability does tell you: can this interface be operated without seeing it, without hearing it, without fine motor control, without holding several items in working memory, without speaking? — which are questions about the interface rather than about any person.

Which makes them testable, by you, today — unplug the mouse; switch the screen reader on; mute the sound; set the zoom to 400% — and that is why the course is organised this way rather than by condition.

And it generalises beyond disability, which is Module 02's subject: 'cannot see the screen right now' covers blindness, bright sunlight, a cracked display, and driving — and the design requirement is identical in all four cases, which is the most useful fact in this module.

Capability	The question to ask of the interface	Module
Vision	Can it be operated without seeing it, and is it usable with low vision or at 400% zoom?	03
Hearing	Is any information available only as audio?	04
Motor control	Is everything reachable by keyboard, and does anything require precision or speed?	05
Cognition and attention	What does it demand of memory, attention, and reading?	06
Speech	Is voice the only way in, anywhere?	04 §4
Photosensitivity and vestibular	Does anything flash, or move large areas, without a way to stop it?	11 §3

Each of these has a test you can run in minutes — which is what makes the list a working checklist rather than a statement of values, and is the difference between this course and an exhortation. Organising by capability rather than by diagnosis is the structural decision of the course, and it is also what lets Module 02 extend the same requirements to temporary and situational cases without changing any of them.

4 The semester

ALL THREE SEMESTER 11 COURSES REPLACE A JUDGEMENT ABOUT TASTE WITH A MEASUREMENT

CSCE 671 replaces 'this feels intuitive' with five participants and a task; CSCE 679 replaces 'this chart looks good' with a measured ranking of the visual channels; and this course replaces 'this seems accessible' with the actual assistive technology and a published standard.

Which is the same move three times — applied to the interaction, to the encoding, and to the access — and it is what makes these three engineering courses rather than criticism, which is the claim CSCE 671 Module 13 §3 and CSCE 679 Module 13 §3 both make from their own side.

And all three correct the same error: designing for an imagined user who resembles the designer — which is CSCE 671 Module 01 §2's designer blindness, and is the failure mode all three semesters' courses are organised against.

So the question this course keeps asking — 'tested with whom, against what?' — is CSCE 671's question with the population widened, and Module 13 is where it gets answered properly, including what a conformance claim does and does not establish.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Microsoft Inclusive Design (free)	§2 and §3 — the clearest short statement of the capability framing, and Module 02 builds directly on it.
Shakespeare — Disability Rights and Wrongs Revisited	§1 argued carefully, including the criticisms of the social model — which are worth knowing. Library copy.
The W3C's Introduction to Web Accessibility (free)	§3's capabilities, with the assistive technologies named and short videos of each in use.
Blackwell — The curb cut effect and its limits (free)	§2's argument in its general form, and useful for Module 12's organisational case.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 State the two models and what each implies about who owns a fix.
- 2 Find a product with a separate accessible version and compare its feature set to the main one.
- 3 List four software curb cuts you personally benefit from.
- 4 Then name two requirements with no mainstream beneficiary, and say why they are still required.

- 5 **Unplug your mouse for an hour** and record what you could not do.
- 6 **Turn on your platform's screen reader** for ten minutes without looking at the screen.
- 7 **Set your browser zoom to 400%** on three sites you use daily.
- 8 **Rewrite one diagnosis-framed requirement** as a capability one.
- 9 **For each capability in §3's table, name the situational version.**
- 10 **State the measurement each Semester 11 course substitutes** for a judgement about taste.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State both models and where each locates the problem.
- 2 Why does the social model suit an engineering course?
- 3 What follows under each model when an interface excludes somebody?
- 4 Why does the separate version lose, and is that a prediction?
- 5 State the curb-cut argument with three software examples.
- 6 State its limit, and why the limit matters.
- 7 Why design against capabilities rather than diagnoses?
- 8 Name six capabilities and the test for each.
- 9 How does the capability framing generalise beyond disability?
- 10 What single error do all three Semester 11 courses correct?